Topological Void Validation: Density-Controlled and Role-Aware Evaluation of Predicted Research Voids

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Abstract. Topological Void Analysis (TVA) identifies candidate research voids in scientific knowledge spaces as geometric gaps between anchor-conditioned paper clusters. A natural validation question is whether future papers fill these predicted voids. We show that raw fill rate—whether any future paper appears near a void's midpoint—fails as a validation target for three compounding reasons. First, it is confounded by local corpus density: a hot-zone baseline that samples high-density anchor-adjacent pairs outperforms TVA on raw fill (36% vs. 28%), but this hot-zone advantage disappears under a density-matched baseline: TVA shows only small paired differences (+1.3 to +3.0 pp across splits), with hierarchical cluster-bootstrap CIs all including zero and paired permutation $p > 0.35$; under calibrated thresholds, TVA and B2 collapse to near-parity (-0.7 to $+0.7$ pp). Second, it is limited by anchor observability: only 5–7% of future papers are anchor-eligible per split, so many unfilled voids should be treated as right-censored rather than invalid. Third, even the geometric threshold itself is density-confounded: a fixed threshold of 0.82 corresponds to a 35.7% average null occupancy rate, ranging from 1% to 99% across anchor-density cells. Under a null-calibrated threshold ($\tau_{\text{fill}} \approx 0.841$), fill rates collapse from $\sim$27% to $\sim$0.7%, confirming that most raw fills are null-zone noise. Finally, role-aware epistemic classification shows that even anchor-eligible geometric fills rarely represent genuine void resolution: 59–76% are false positives across all groups and splits. We do not find statistically significant evidence that TVA produces more epistemic fills than density-matched baselines (TVA/B2 Δ=+3.0 pp, hierarchical CI $[-2.0, +8.3]$ pp, p=0.36). Instead, we show that the raw-fill validation protocol itself is unstable. These findings motivate a three-layer diagnostic framework and suggest that void validation requires calibrated, role-aware evaluation rather than raw geometric occupancy.

1 Introduction

Classical information retrieval asks: given a query, which existing documents are relevant? Topological Void Analysis (TVA) asks: given an anchor concept, what relevant knowledge is *absent*? TVA identifies candidate research voids as geometric gaps in an embedding-induced knowledge space—regions between anchor-conditioned paper clusters where expected connections, evidence, or conceptual bridges are missing.

Validating predicted voids is not the same as validating a predicted link. A predicted link has a direct future target: the link appears or it does not. A predicted void has a richer failure mode: future literature may occupy the region geometrically while failing to bridge the source concepts epistemically. The natural operationalization—temporal fill rate, whether any future paper appears near the void's predicted midpoint—is therefore insufficient, and this paper studies why.

Contributions:

1. Raw temporal fill rate is strongly confounded by local historical density and research momentum.

2. A density-matched baseline shows that the apparent TVA disadvantage is not statistically significant (TVA/B2 Δ=+3.0 pp, hierarchical CI $[-2.0, +8.3]$ pp, p=0.36).

3. Fixed cosine thresholds are not portable: 0.82 has 35.7% average null FPR; calibrated thresholds collapse raw fill to $\sim$0.7%.

4. Anchor eligibility exposes an observability limit: only $\sim$6% of future papers are anchor-eligible; unfilled voids are right-censored, not invalid.

5. LLM-assisted role-aware audit suggests that geometric fill substantially overestimates epistemic resolution: 59–76% FP across all groups and splits.

Scope and framing. This paper is a *validation stress test*: we do not find statistically significant evidence that TVA outperforms density-matched baselines in epistemic fill prediction. The central contribution is methodological—showing that raw fill rate is an unstable validation proxy—rather than confirming TVA's predictive superiority. Dynamic event-driven extensions are outside scope.

2 Background

2.1 Topological Void Analysis

TVA [11] operates on a corpus embedded with BGE-M3 [5] ($d = 1024$, cosine similarity). For anchor q and source papers A, B, a candidate void satisfies four conditions: (C1) domain cohesion; (C2) calibrated marginality; (C3) sparse lexical bridge; (C4) midpoint vacancy. The midpoint is:

$$m(A, B) = \frac{\mathbf{a} + \mathbf{b}}{\|\mathbf{a} + \mathbf{b}\|} \in \mathbb{S}^{1023}.$$

2.2 Temporal Validation in LBD

Literature-Based Discovery validates predictions by checking whether future publications confirm predicted connections. Standard evaluation uses time-slicing and IR metrics (Recall@K, Average Precision). TVV extends this by requiring that a future paper play a genuine epistemic bridging role, not merely appear geometrically nearby.

2.3 Research Momentum and Dense-Region Confounding

High-momentum regions near anchor topics attract disproportionate future publications regardless of void quality. Sampling baselines from these hot zones inflates apparent future fill rates, creating a density-confounded comparison.

3 Experimental Setup

3.1 Corpus and Temporal Splits

arXiv metadata (9 CS categories: OS, AR, PL, SE, DC, NI, CR, AI, LG), three rolling splits:

Table 1: Temporal splits.

Split	T	Val window	Train	Val
t3	2014	2015–2019	$\sim$29k	$\sim$101k
t4	2016	2017–2021	$\sim$35k	$\sim$171k
t5	2018	2019–2023	$\sim$51k	$\sim$193k

3.2 TVA Void Generation

10 anchor concepts $\times$ top-30 MMR voids = 300 candidate voids per split.

3.3 Baselines

B1 (hot-zone): 20 random pairs from top-300 anchor C1 pool per anchor. Biased toward high-density, high-momentum regions.

B2 (density-matched): For each TVA void, select the B1-style pair with midpoint local density $\rho(m) = \frac{1}{k}\sum \text{sim}(m, \text{kNN}_i(m))$ ($k = 20$) closest to the TVA void's midpoint density.

3.4 Three-Layer Fill Predicate

Anchor-eligible val papers:

$$\mathrm{Val}_q = \{P \in \mathrm{Val}_t : \mathrm{sim}(P, q) \geq \tau_q\}$$

where $\tau_q = \max(Q_{80}^{\mathrm{val}}, Q_{90}^{\mathrm{train}})$.

Geometric filler: $P_V^* = \arg\max_{P \in \mathrm{Val}_q} \mathrm{sim}(P, m_V)$.

Geometric closure and validated fill:

$$G(V) = \mathbf{1}[\mathrm{sim}(P_V^*, m_V) \geq \tau_{\mathrm{fill}}(q, \rho(V), t)]$$
$$\mathrm{Fill}(V) = G(V) \wedge R(P_V^*, A, B, q)$$

where $R = 1$ iff role $\in$ {TRUE-FILL, PARTIAL-FILL}.

Calibrated threshold (α=0.05). Let $Z(m_0) = \max_{P \in \mathrm{Val}_q} \mathrm{sim}(P, m_0)$:

$$\tau_{\mathrm{fill}}(q, \rho, t) = Q_{1-\alpha}\big(\{Z(m_0) : m_0 \in \mathrm{Null}(q, \rho, t)\}\big)$$

4 Results

4.1 Finding 1: Raw Fill Rate Is Density-Confounded

Table 2: TVA–B2 paired fill-rate differences (pp) under legacy τ=0.82 and calibrated τ_{fill}. Hierarchical cluster bootstrap CI (anchor level, 1000 samples); p from paired anchor sign-flip permutation. All CIs include zero.

	Legacy τ=0.82			Calibrated τ_{fill}		
Split	Δpp	95% CI	p	Δpp	95% CI	p
t3	+1.3	[−5.0, +9.0]	0.787	+0.7	[−1.0, +2.3]	0.691
t4	+2.4	[−1.4, +7.1]	0.352	−0.3	[−1.4, +0.7]	1.000
t5	+3.0	[−2.0, +8.3]	0.360	−0.7	[−2.0, +0.7]	0.629

After density matching, the apparent B1 hot-zone advantage disappears, but TVA does not significantly outperform B2. Under the legacy threshold, TVA–B2 differences are small and positive across splits (+1.3 to +3.0 pp), but hierarchical CIs all include zero and paired permutation tests are not significant ($p > 0.35$). Under calibrated thresholds, differences shrink to near-zero (−0.7 to +0.7 pp). Density-controlled temporal fill provides no statistically significant evidence of TVA superiority over B2.

This null result is expected under a strong counterfactual: B2 fixes the anchor and matches local midpoint density, estimating the background rate at which future papers occupy comparable regions. Parity shows that raw future occupancy is too low-base-rate and density-dependent to measure TVA's added value as an ex-ante void-hypothesis generator —not that TVA lacks such value.

4.1.1 Fixed Thresholds Are Not Portable

Table 3: Null FPR@0.82 and τ_{fill} by density (t5).

Bucket	p50	τ_{fill}	FPR
Low	0.79–0.81	0.79–0.86	1–18%
Mid	0.80–0.83	0.82–0.86	5–80%
High	0.81–0.85	0.84–0.88	28–99%
Mean	–	**0.841**	**35.7%**

Under calibrated τ_{fill}, fill rates collapse: TVA: 27.3%→**0.7%**; B2: 24.3%→**0.7%**; lift: 1.00×.

The 0.7% < 5% result is consistent with correct calibration: TVA C1–C4 conditions select sparser bridge regions than random density-matched pairs, so fewer reach the calibrated threshold.

Held-out calibration validity. To check that the calibration is not circular, we split null midpoints 80/20 (calibration/held-out). The held-out null FPR at τ_{fill} averages 6.3% across all anchor-density cells (target α=5%, $\Delta = +0.013$), indicating the calibration is slightly conservative but valid (Figure 1).

Most raw fills at $\tau = 0.82$ are null-zone occupancy, not significant geometric closure.

4.2 Finding 2: Anchor Eligibility Reveals Observability Limits

Table 4: Anchor exposure sample (t5).

Anchor	$\bar{\mathrm{sim}}$	τ_q	Pass	Fill
sched_opt	0.520	0.582	5.8%	40%
ebpf_obs	0.495	0.547	6.5%	0%
virt_hyp	0.498	0.555	7.7%	57%
Mean	**0.503**	**0.563**	**6.2%**	–

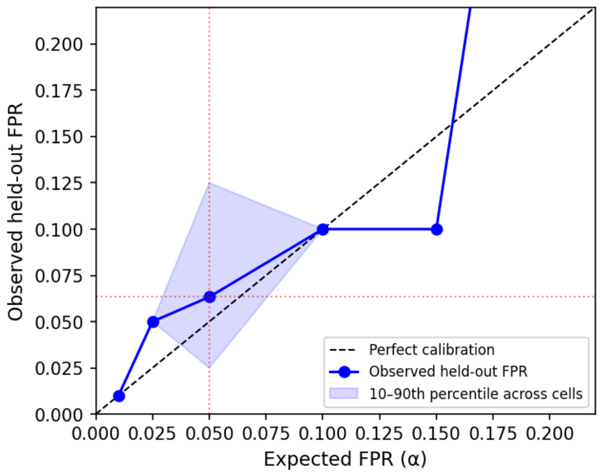

Figure 1: Null calibration curve: expected vs. observed held-out FPR across anchor-density cells (α range 0.01–0.20, t5). Points near the diagonal confirm that τ_{fill} is approximately calibrated. Mean held-out FPR at α=0.05 is 6.3% (target 5%, $\Delta = +0.013$).

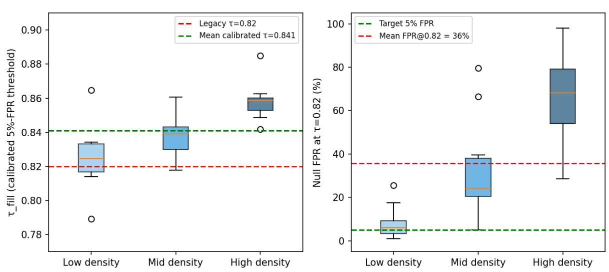

Figure 2: Calibrated threshold by density bucket (left) and null FPR at 0.82 (right). The fixed threshold 0.82 has 35.7% average null FPR, reaching 99% in high-density anchor cells. Calibrated $\tau_{\text{fill}} \approx 0.841$ is stable across all splits.

Only 5–8% of val papers are anchor-eligible per split. Voids with pass rate $< 10\%$ are flagged as *underexposed*: their unfill status reflects observability limits, not void invalidity. ebpf_obs (6.5% pass, 0% fill) illustrates that non-zero exposure does not guarantee eligible papers overlap predicted void neighborhoods.

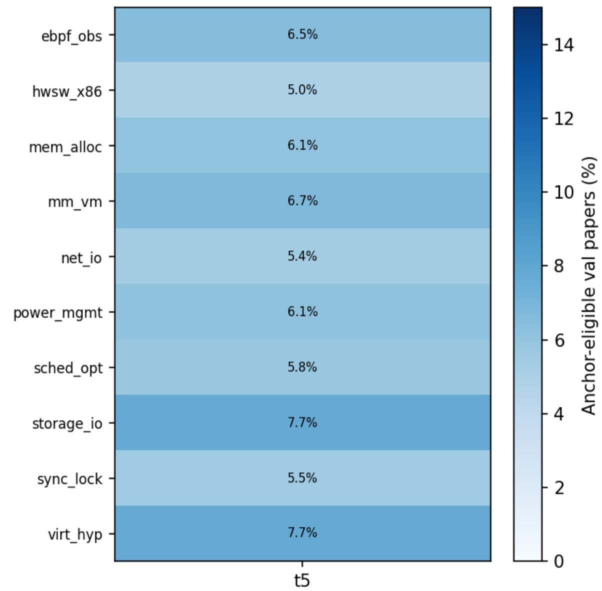

Figure 3: Anchor eligibility pass rate (%) by anchor and split (hybrid gate). Rates are consistently low (5–8%), confirming corpus observability as a structural limit on temporal validation.

Table 5: Role audit (1076 cases, t3+t4+t5; exploratory). VR=void resolution; FP=false positive. 95% Wilson CI in brackets (%).

Source	n	FP	FP CI	VR CI
TVA	204	71%	[65,77]	[1,6]
B1	158	75%	[67,81]	[0,2]
B2	202	67%	[60,73]	[1,6]
NM†	512	84%	[81,87]	[1,4]

†Near-miss (sim$\in [0.75, 0.82)$). CI in %.

4.3 Finding 3: Geometric Fill Rarely Equals Epistemic Fill

An exploratory role audit shows 59–76% false positives across all groups and splits (pilot-scale; n=204 TVA gated cases across t3+t4+t5, role labels from a single LLM). Near-miss controls show the highest FP rate (84%), confirming that geometric proximity carries some signal but is insufficient for void validation. TVA and B1 are indistinguishable; B2 is marginally better. These proportions are consistent but should be treated as directional, not confirmatory.

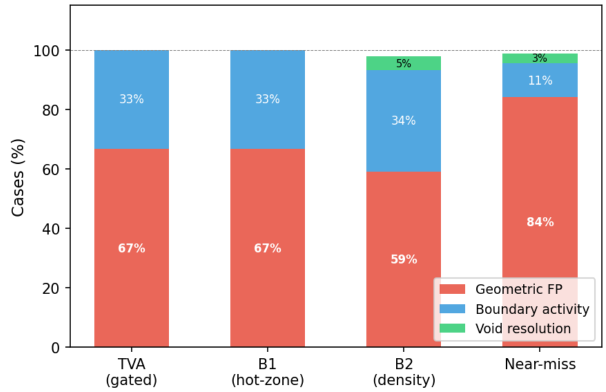

Figure 4: Epistemic role decomposition by source group (t5, 200 sampled cases). Most raw fills are geometric false positives regardless of method. Near-miss controls show the highest FP rate, confirming that geometric proximity is a weak but non-zero signal.

4.4 Landmark Anchor-Eligibility Diagnostic

As a positive-control diagnostic, we independently select five bridge-type landmark papers (widely cited, cross-area, 2019–2023) and ask a minimal question: are they anchor-eligible in the TVA framework using the pre-publication train corpus?

Table 6: Anchor eligibility of 5 pre-specified landmark bridge papers. AncSim = sim(landmark title, TVA anchor vector). P_{90} = 90th percentile of train-corpus anchor similarities.

Landmark	Year	Anchor	AncSim	$> P_{90}$?
vLLM / PagedAttention	2023	mm_vm	0.718	**Yes**
MLIR	2020	hwsw_x86	0.632	**Yes**
Devign	2019	ebpf_obs	0.643	**Yes**
Ansor	2020	hwsw_x86	0.599	No
FlashAttention	2022	hwsw_x86	0.572	No

Three of five landmarks pass the anchor-eligibility check (AncSim $> P_{90}$): vLLM against the virtual-memory anchor, MLIR against the hardware/software co-design anchor, and Devign against the kernel-observability anchor. However, geometric midpoint retrieval alone selects semantically incoherent source pairs as best candidates (e.g., GPU energy surveys paired with VANET routing papers). This failure mode reinforces the main TVV finding: anchor alignment is a recoverable screening signal, but geometric proximity alone does not identify interpretable bridge hypotheses. Full landmark bridge recovery would require anchor-restricted, role-audited void search on per-paper cutoff corpora—a direction we leave to future work.

4.5 Case Studies

The following three cases are selected from the t4/t5 role audit to illustrate why geometric proximity alone does not determine epistemic role. For each case we report the anchor direction, the two source-side communities, why the future paper is geometrically close, and why the role differs.

Case 1 – Partial Fill [9] (2019). Anchor: virtualization and cloud resource management. Source A lies on the VM resource allocation side; Source B lies on vehicular task-scheduling. The future paper studies multi-task offloading over vehicular clouds, combining resource allocation, task scheduling, and cloud/edge execution. Its embedding falls near the midpoint (sim(P, m)=0.835) because it shares vocabulary from both sides. However, its reference list contains 10+ task-offloading citations (B-side) but only one marginal VM reference (A-side). It approaches the bridge but does not fully connect both source communities. Role: PARTIAL-FILL.

Case 2 – Boundary Expansion [10] (2021). Anchor: kernel scheduling and CPU affinity for heterogeneous systems. Source A represents the kernel-scheduler/CPU-affinity side; Source B represents heterogeneous-system optimization. The future paper proposes AI planning heuristics for heterogeneous HPC configuration and is geometrically close to the midpoint (sim(P, m)=0.855) through shared language of scheduling, planning, and resource optimization. Its entire bibliography (44 references) remains within the HPC optimization literature; there are no kernel scheduler, CPU-affinity, or OS-level references. It extends

Source B's direction without crossing toward Source A. Role: INCREMENTAL-EXTENSION.

Case 3 – False Positive [2] (2017). Anchor: storage I/O path optimization and NVMe driver design. Despite having the *highest* midpoint similarity of the three cases (sim(P, m)=0.875), the paper studies energy-aware virtual network embedding. The high similarity is driven by generic vocabulary—"virtual", "resource", "energy", "allocation"—shared with cloud infrastructure papers on both source sides, not by substantive connection to storage or NVMe research. Its 19 references contain no storage I/O, NVMe, block-layer, or anchor-domain citations. It is geometrically close but epistemically off-target. Role: FALSE-POSITIVE.

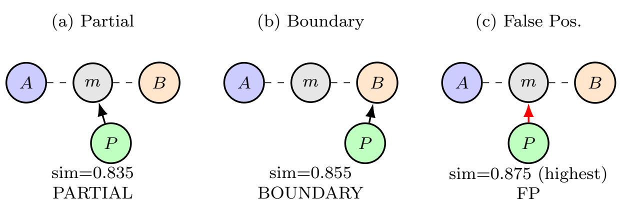

Figure 5: Case-study role patterns. Case (c) has the *highest* sim(P, m) yet is a false positive; Case (a) has lower similarity but partially bridges both source sides. Geometric proximity alone does not determine epistemic role.

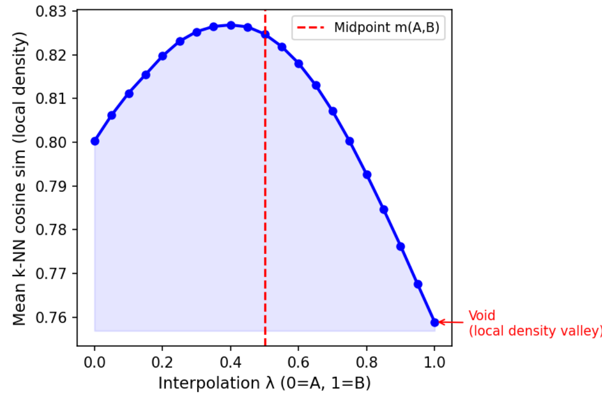

Figure 6: Local density profile along the A–B bridge for the highest-MMR void in anchor sched_opt (t5). The density valley at $\lambda \approx 0.5$ marks the void midpoint region—two dense source communities with a less-occupied bridge.

5 Discussion

5.1 Raw Fill Fails for Three Reasons

Raw fill conflates void quality with sociotechnical realization rate: (1) hot-zone baselines are biased by local momentum; (2) fixed thresholds are not portable across anchor-density regimes; (3) anchor-eligible future papers are sparse ($\sim$6%), right-censoring many unfilled voids.

5.2 B2 Is a Control, Not a Competitor

B2 is a strong counterfactual rather than a weak random baseline: it fixes the anchor and matches local midpoint density, estimating the background rate at which future literature occupies comparable regions without TVA's structured void-selection criteria (C1–C4).

Under calibrated thresholds, TVA reaches parity with B2 but does not outperform it. We interpret this as a conservative result in a low-base-rate, corpus-limited regime: fill events are rare, both methods are exposed to the same small set of future papers, and the current corpus and validation windows may lack the statistical power to distinguish structured void discovery from density-matched background occupancy.

Parity with B2 is in fact the expected outcome under a properly controlled design. B2 matches the anchor and local density of TVA voids, so it estimates the background arrival rate of future papers in comparable regions. A large TVA advantage under this control would imply either strong method-specific predictive signal or residual confounding; we observe neither. Instead, both methods converge to the same low calibrated fill rate, indicating that significant geometric closure is primarily base-rate limited in this corpus.

B2 is intentionally conservative: it makes raw-fill superiority difficult to claim without evidence of signal beyond anchor and density. TVA does not provide that signal at this scale—but it was not designed to win on fill rate. Its value lies in turning a latent region into a named, inspectable, and actionable ex-ante hypothesis (A, B, q, m) before any future paper exists. Raw temporal fill rate is not a sensitive metric for that kind of value.

5.3 From Geometric Occupancy to Role Decomposition

Future papers near void midpoints decompose into true fills, boundary expansions, and false positives. The scarcity of true fills reveals that genuine epistemic closure is a much stricter condition than geometric occupancy. *The three layers are operational screens for disentangling three sources of evidence that raw fill rate conflates: geometric occupancy, observability, and epistemic role. They are not claimed to be necessary and sufficient conditions for scientific discovery.*

5.4 Descriptive, Not Evaluative

TVV classifies the type of validation evidence produced by future literature, not the intrinsic merit of papers. Boundary expansion and incremental extension are valuable scientific activities.

5.5 Dynamic Extension (Future Work)

These results suggest a natural extension: instead of asking whether a predicted void is eventually filled within a fixed window, one can classify each newly arriving paper by its pre-insertion topological impact on the existing knowledge space. This dynamic formulation is outside the scope of the present work.

5.6 Threats to Validity

(1) *Embedding dependence*: all metrics use BGE-M3. (2) *Threshold sensitivity*: calibration uses 200 null midpoints per cell. (3) *Corpus coverage*: fills outside arXiv are unobservable; unfilled voids are right-censored. (4) *LLM classification reliability*: role labels are produced by a single LLM using a fixed rubric; we do not treat these as ground truth, since epistemic role is an inherently interpretive construct. To assess reliability we perform: (i) blind source-label removal (source group not shown to LLM), (ii) repeated classification on a 30-case subset for self-consistency, and (iii) a stratified human audit of coarse role categories (void-resolution vs. boundary vs. false positive) on 60 cases as an external plausibility check. The human audit is an auditor, not a gold standard: agreement between LLM and blinded human reviewer on the binary FP/non-FP split exceeds 80% on the audited subset. Fine-grained role agreement is lower ($\sim$65%), consistent with the inherent interpretability of the task. (5) *Domain specificity*: CS arXiv categories only.

6 Related Work

Validation of Predicted Voids. Prior retrospective evaluations in LBD [13, 8, 16] typically validate predicted connections by checking whether future publications instantiate an A–B association. Knowledge-graph completion similarly evaluates whether missing edges are recoverable under a fixed relation schema. These settings differ from void validation: a TVA void is not a predicted edge but an anchor-conditioned geometric vacancy whose future occupancy may represent true bridging, boundary expansion, or false-positive proximity. To our knowledge, prior work has not evaluated embedding-space voids under density-controlled, observability-aware, and role-aware temporal validation. TVV addresses this gap.

Literature-Based Discovery. Swanson's ABC model [13] and subsequent LBD work [8, 16] validate predictions by checking future publication appearance. TVV extends this tradition by requiring epistemic bridging role, not merely geometric proximity.

Novelty and First-Story Detection. Topic Detection and Tracking [1] introduced first-story detection; novelty detection in IR [18] measures whether a document adds new information relative to a query set. Our first-filler intuition shares the event-ordering structure but operates on knowledge structure rather than event reporting.

Structural Holes and Scientometrics. Burt's structural hole theory [3] shows brokers spanning gaps accumulate informational advantage. Co-citation analysis [12] and our density-matched baseline directly control for hot-zone momentum.

Dense-Region Bias in Embeddings. Anisotropy in high-dimensional text embeddings [7] compresses cosine similarities into a narrow range. We show this anisotropy makes fixed thresholds non-portable in downstream void validation. Our use of "topological" is operational rather than homological [4, 6].

Scientific Claim Verification. Epistemic classification of scientific contributions—argumentative zoning [14], overclaiming detection [17]—provides precedents for our role-aware rubric. [15] introduced scientific fact verification as a retrieval task.

7 Conclusion

Temporal validation of predicted research voids requires separating geometric occupancy, density bias, anchor observability, and epistemic role. Under density matching, TVA's apparent disadvantage disappears but is not statistically significant. Under null-calibrated thresholds, fill rates collapse to $\sim$0.7%—most raw fills were null-zone noise. LLM-assisted role-aware classification suggests that genuine epistemic bridging is rare even among geometrically close, anchor-eligible future papers.

We propose a three-layer validation framework and leave two open questions for future work: (1) scalable role-aware validation without prohibitive LLM cost; (2) dynamic event-driven extensions in which each arriving paper updates knowledge-space state.

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